Machine Learning Lab – Question 14

Question

Implement House price Prediction using an appropriate machine learning algorithm.

Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score

data=pd.DataFrame({
    "area":[800,1000,1200,1500,1800,2000,2200,2500,2800,3000],
    "bedrooms":[2,2,3,3,3,4,4,4,5,5],
    "age":[20,15,12,10,8,6,5,4,3,2],
    "price":[35,42,50,65,78,90,105,120,140,155]
})
X=data[["area","bedrooms","age"]]; y=data["price"]
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=.2,random_state=42)

model=RandomForestRegressor(n_estimators=200,random_state=42)
model.fit(X_train,y_train)
pred=model.predict(X_test)

print("MAE:",round(mean_absolute_error(y_test,pred),2))
print("R2:",round(r2_score(y_test,pred),4))
print("Predicted price:",round(model.predict([[1800,3,5]])[0],2),"lakhs")
```

Output

```
The program prints MAE, R2 score, and a predicted house price.
Exact values depend on the sample split and model version.
```